Lab 6 - Batoul Hachem - pybuilder

```
LAB 6 - AUTOMATIC BUILD | Batoul Hachem
Project: C:\Users\User\Downloads\calculator_Batoul_Hachem

PS> pyb run_unit_tests
PyBuilder version 0.13.24
Build started at 2026-10-26 14:19:56
------------------------------------------------------------
[INFO]  Building calculator_Batoul_Hachem version 0.1
[INFO]  Executing build in c:\users\user\downloads\calculator_batoul_hachem
[INFO]  Going to execute task run_unit_tests
[INFO]  Processing plugin packages 'build>=1.3.0' to be installed with {}
[INFO]  Processing plugin packages 'pypandoc~=1.4' to be installed with {'upgrade': True}
[INFO]  Processing plugin packages 'setuptools>=76.0' to be installed with {}
[INFO]  Processing plugin packages 'twine>=1.15.0' to be installed with {'upgrade': True}
[INFO]  Processing plugin packages 'unittest-xml-reporting>=3.0.4' to be installed with {'upgrade': True}
[INFO]  Creating target 'build' VEnv in 'c:\users\user\downloads\calculator_batoul_hachem\target\venv\build\cpython-3.13.2.final.0'
[INFO]  Creating target 'test' VEnv in 'c:\users\user\downloads\calculator_batoul_hachem\target\venv\test\cpython-3.13.2.final.0'
[INFO]  Running unit tests
[INFO]  Executing unit tests from Python modules in c:\users\user\downloads\calculator_batoul_hachem\tests\unittest\calculator_batoul_hachem
[INFO]  Executed 21 unit tests
[INFO]  All unit tests passed.
------------------------------------------------------------
BUILD SUCCESSFUL
------------------------------------------------------------
Build Summary
            Project: calculator_Batoul_Hachem
            Version: 0.1
     Base directory: c:\users\user\downloads\calculator_batoul_hachem
       Environments:
              Tasks: prepare [7294 ms] compile_sources [0 ms] run_unit_tests [325 ms]
Build finished at 2026-10-26 14:20:06
Build took 9 seconds (9524 ms)

PS>
```

Lab 6 - Batoul Hachem - unittest

```
LAB 6 - AUTOMATIC BUILD | Batoul Hachem
Project: C:\Users\User\Downloads\calculator_Batoul_Hachem

PS> python -m unittest discover -s tests/unittest/calculator_Batoul_Hachem -v
test_add (test_calculator_Batoul_Hachem.TestCalculator) ... ok
test_add_negative (test_calculator_Batoul_Hachem.TestCalculator) ... ok
test_divide (test_calculator_Batoul_Hachem.TestCalculator) ... ok
test_divide_by_zero (test_calculator_Batoul_Hachem.TestCalculator) ... ok
test_divide_fraction (test_calculator_Batoul_Hachem.TestCalculator) ... ok
test_modulus (test_calculator_Batoul_Hachem.TestCalculator) ... ok
test_modulus_by_zero (test_calculator_Batoul_Hachem.TestCalculator) ... ok
test_modulus_exact_multiple (test_calculator_Batoul_Hachem.TestCalculator) ... ok
test_modulus_negative (test_calculator_Batoul_Hachem.TestCalculator) ... ok
test_multiply (test_calculator_Batoul_Hachem.TestCalculator) ... ok
test_multiply_by_zero (test_calculator_Batoul_Hachem.TestCalculator) ... ok
test_power (test_calculator_Batoul_Hachem.TestCalculator) ... ok
test_power_negative_base (test_calculator_Batoul_Hachem.TestCalculator) ... ok
test_power_negative_exponent (test_calculator_Batoul_Hachem.TestCalculator) ... ok
test_power_zero_exponent (test_calculator_Batoul_Hachem.TestCalculator) ... ok
test_square_root (test_calculator_Batoul_Hachem.TestCalculator) ... ok
test_square_root_negative (test_calculator_Batoul_Hachem.TestCalculator) ... ok
test_square_root_nonperfect_square (test_calculator_Batoul_Hachem.TestCalculator) ... ok
test_square_root_zero (test_calculator_Batoul_Hachem.TestCalculator) ... ok
test_subtract (test_calculator_Batoul_Hachem.TestCalculator) ... ok
test_subtract_negative_result (test_calculator_Batoul_Hachem.TestCalculator) ... ok

----------------------------------------------------------------------
Ran 21 tests in 0.005s

OK

PS>
```

Lab 6 - Batoul Hachem - make

```
LAB 6 - AUTOMATIC BUILD | Batoul Hachem
Project: C:\Users\User\Downloads\calculator_Batoul_Hachem

PS> make build
pyb publish
PyBuilder version 0.13.24
Build started at 2026-10-26 14:20:09
------------------------------------------------------------
[INFO]  Building calculator_Batoul_Hachem version 0.1
[INFO]  Executing build in c:\users\user\downloads\calculator_batoul_hachem
[INFO]  Going to execute task publish
[INFO]  Processing plugin packages 'build>=1.3.0' to be installed with {}
[INFO]  Processing plugin packages 'pypandoc~=1.4' to be installed with {'upgrade': True}
[INFO]  Processing plugin packages 'setuptools>=76.0' to be installed with {}
[INFO]  Processing plugin packages 'twine>=1.15.0' to be installed with {'upgrade': True}
[INFO]  Processing plugin packages 'unittest-xml-reporting>=3.0.4' to be installed with {'upgrade': True}
[INFO]  Creating target 'build' VEnv in 'c:\users\user\downloads\calculator_batoul_hachem\target\venv\build\cpython-3.13.2.final.0'
[INFO]  Creating target 'test' VEnv in 'c:\users\user\downloads\calculator_batoul_hachem\target\venv\test\cpython-3.13.2.final.0'
[INFO]  Running unit tests
[INFO]  Executing unit tests from Python modules in c:\users\user\downloads\calculator_batoul_hachem\tests\unittest\calculator_batoul_hachem
[INFO]  Executed 21 unit tests
[INFO]  All unit tests passed.
[INFO]  Building distribution in c:\users\user\downloads\calculator_batoul_hachem\target\dist\calculator_batoul_hachem-0.1
[INFO]  Copying scripts to c:\users\user\downloads\calculator_batoul_hachem\target\dist\calculator_batoul_hachem-0.1\scripts
[INFO]  Writing setup.py as c:\users\user\downloads\calculator_batoul_hachem\target\dist\calculator_batoul_hachem-0.1\setup.py
[INFO]  Building binary distribution in c:\users\user\downloads\calculator_batoul_hachem\target\dist\calculator_batoul_hachem-0.1
[INFO]  Running Twine check for generated artifacts
------------------------------------------------------------
BUILD SUCCESSFUL
------------------------------------------------------------
Build Summary
            Project: calculator_Batoul_Hachem
            Version: 0.1
     Base directory: c:\users\user\downloads\calculator_batoul_hachem
       Environments:
              Tasks: prepare [7405 ms] compile_sources [0 ms] run_unit_tests [323 ms] package [6 ms] run_integration_tests [0 ms] verify [0 ms] publish [9910 ms]
Build finished at 2026-10-26 14:20:29
Build took 19 seconds (19613 ms)
```

Lab 6 - Batoul Hachem - files

```
LAB 6 - AUTOMATIC BUILD | Batoul Hachem
Project: C:\Users\User\Downloads\calculator_Batoul_Hachem

PS> Get-ChildItem src,tests -Recurse -Filter *.py
src\main\python\calculator_Batoul_Hachem\__init__.py
tests\__init__.py
tests\unittest\__init__.py
tests\unittest\calculator_Batoul_Hachem\test_calculator_Batoul_Hachem.py
tests\unittest\calculator_Batoul_Hachem\__init__.py

PS> Get-ChildItem target/dist/*/dist

Name                                           Length
----                                           ------
calculator_batoul_hachem-0.1-py3-none-any.whl    3601
calculator_batoul_hachem-0.1.tar.gz              3375



PS>
```